Q7(a)

$$\frac{Q}{t} = kA\frac{\theta_1 - \theta_2}{\Delta x}$$

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$$k = \frac{Q\Delta x}{tA(\ \theta_1 - \theta_2\ )}$$

Base units of k = $\frac{(\text{kg m}^2\text{s}^{-2})(\text{m})}{(\text{s})(\text{m}^2)(\text{K})}$

= **kg m s^{-3} K^{-1}**

7(b)

The negative sign is used to show that the temperature θ decreases in the direction of *x*.

7(c)(i)

Mass of water $M = \rho V$

$=\rho(\pi {r_1}^2 L)$

= (1000)(π) (4.0x10^{-2})^2 (10.0x10^{-2})

= **0.503 kg**

7(c)(ii)

Total surface area would include the top and bottom of the mug.

$A = 2\pi {r_1}^2 + 2\pi r_1 L$

= (2π)(4.0x10^{-2})^2 + (2π)(4.0x10^{-2})(10.0x10^{-2})

≈ **0.0352 m^2**

7(c)(iii)

The rate of loss of energy by the tea water = rate of energy lost through the walls and top and bottom of the mug.

$\frac{Q}{t} = kA\frac{\theta_1 - \theta_2}{\Delta x} = \frac{Mc(\Delta T)}{t}$

$$\Delta T = \frac{kAt(\theta_{tea} - \theta_{room})}{(\Delta x)Mc}$$

Hence

$$\Delta T = \frac{(1.0)(0.0352)(30)(\theta_{tea} - \theta_{room})}{(5.0\text{x}10^{-3})(0.503)(4180)}$$

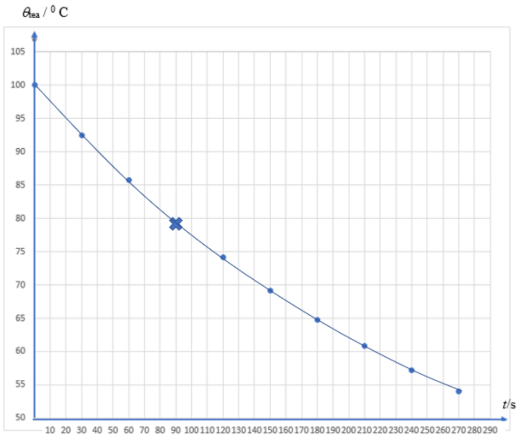
Hence $\boldsymbol{\Delta T = 0.10(\theta_{tea} - \theta_{room})}$

(7d)(i)

Time after pouring *t*/s	**Initial θ_{tea}/ ^{0} C**	**Temperature change in 30 s ΔT/ ^{0} C**	**Final θ_{tea}/ ^{0} C**
0	100	7.5	92.5
30	92.5	6.8	85.7
60	85.7	6.1	79.6
90	**79.6**	**5.5**	**74.1**
120	74.1	4.9	69.2

150	69.2	4.4	64.8
180	64.8	4.0	60.8
210	60.8	3.6	57.2
240	57.2	3.2	54.0
270	54.0	2.9	51.1

7(d)(ii)



[2]

7(d)(iii)
For the tea to drop from 100 ^{0}C to 55 ^{0}C, the time is about 260 s.

7(e)(i)
$A = 2\pi rL = (2\pi)(4.0\text{x}10^{-2})(10.0\text{x}10^{-2})$= 0.0251 m^2

Using $\frac{Q}{t} = kA\frac{\theta_1-\theta_2}{\Delta x}$, we have

$$\frac{Q}{t} = (1.0)(0.0251)\frac{(90-25)}{5.0\text{x}10^{-3}}$$

= **326 W**

7(e)(ii)
Using $\frac{dQ}{dt} = \frac{k2\pi L(\theta_1-\theta_2)}{ln\left(\frac{r_2}{r_1}\right)}$, we have

$\frac{dQ}{dt} = \frac{(1.0)(0.628)(90-25)}{ln\left(\frac{0.045}{0.040}\right)}$ ≈ **347 W**

7(e)(iii)
Percentage error is $\frac{347-326}{347}$ x 100 ≈ **6.1%**

7(f)

$$P = \frac{Mc\Delta T}{t} \text{ or } t = \frac{Mc\Delta T}{P}$$

$$t = \frac{(0.503)(4180)(70 - 55)}{1.1 \text{ x } 10^3}$$

≈28.7 s